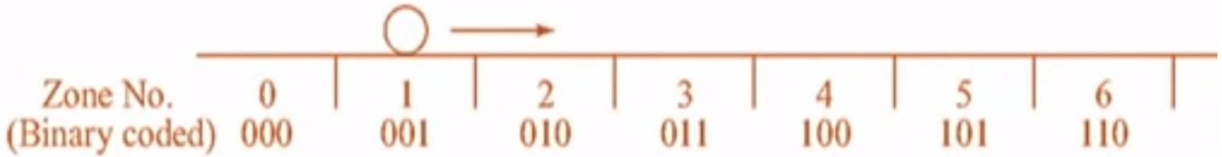
Unweighted Codes

	B_3	B_2	B_1	B_0
	0	0	0	0
2 digits change	0	0	0	1
	0	0	1	0
3 digits change	0	0	1	1
	0	1	0	0
	0	1	0	1
.	0	1	1	0
.	0	1	1	1
	1	0	0	0
.	1	0	0	1
	1	0	1	0



$B_2B_1B_0$:
(If B_1 change is slower than B_0)

001 → 000 → 010

Possibility of decoding a wrong direction of movement

$$d_n \dots d_1 d_0 . d_{-1} d_{-2} \dots d_{-m} = d_n \times r^n + \dots + d_1 \times r^1 + d_0 \times r^0 + d_{-1} \times r^1 + d_{-2} \times r^2 + \dots + d_{-m} \times r^m$$

1010 in 8421 code = 1 x 8 + 0 x 4 + 1 x 2 + 0 x 1

8421 : Weights associated with position

Note: 1010 in 2421 code = 1 x 2 + 0 x 4 + 1 x 2 + 0 x 1

Gray Code

- Unit Distance Code
- Unweighted Code
- Self reflecting code

Self Reflecting Codes

- By reflecting (n-1) bit gray code, n-bit gray code can be obtained

0
1

0	0
0	1
1	1
1	0

0	0 0
0	0 1
0	1 1
0	1 0
1	1 0
1	1 1
1	0 1
1	0 0

0	0 0 0
0	0 0 1
0	0 1 1
0	0 1 0
0	1 1 0
0	1 1 1
0	1 0 1
0	1 0 0
1	1 0 0
1	1 0 1
1	1 1 1
1	1 1 0
1	0 1 0
1	0 1 1
1	0 0 1
1	0 0 0

Binary to Gray Code Conversion

DECIMAL DIGIT	BINARY CODE				GRAY CODE			
	B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0
4	0	1	0	0	0	1	1	0
5	0	1	0	1	0	1	1	1
6	0	1	1	0	0	1	0	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	1	0	0
9	1	0	0	1	1	1	0	1
10	1	0	1	0	1	1	1	1
11	1	0	1	1	1	1	1	0
12	1	1	0	0	1	0	1	0
13	1	1	0	1	1	0	1	1
14	1	1	1	0	1	0	0	1
15	1	1	1	1	1	0	0	0

B_3	B_2	B_1	B_0	G_3
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

B_3B_2 \ B_1B_0	00	01	11	10
00				
01				
11	1	1	1	1
10	1	1	1	1

$$\boxed{G_3 = B_3}$$

B_3	B_2	B_1	B_0	G_2
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

B_3B_2 \ B_1B_0	00	01	11	10
00				
01	1	1	1	1
11				
10	1	1	1	1

$$G_2 = B_3B_2' + B_2B_3'$$

$$\boxed{G_2 = B_3 \oplus B_2}$$

B_3	B_2	B_1	B_0	G_1
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

B_3B_2 \ B_1B_0	00	01	11	10
00			1	1
01	1	1		
11	1	1		
10			1	1

$$G_1 = B_2B_1' + B_1B_2'$$

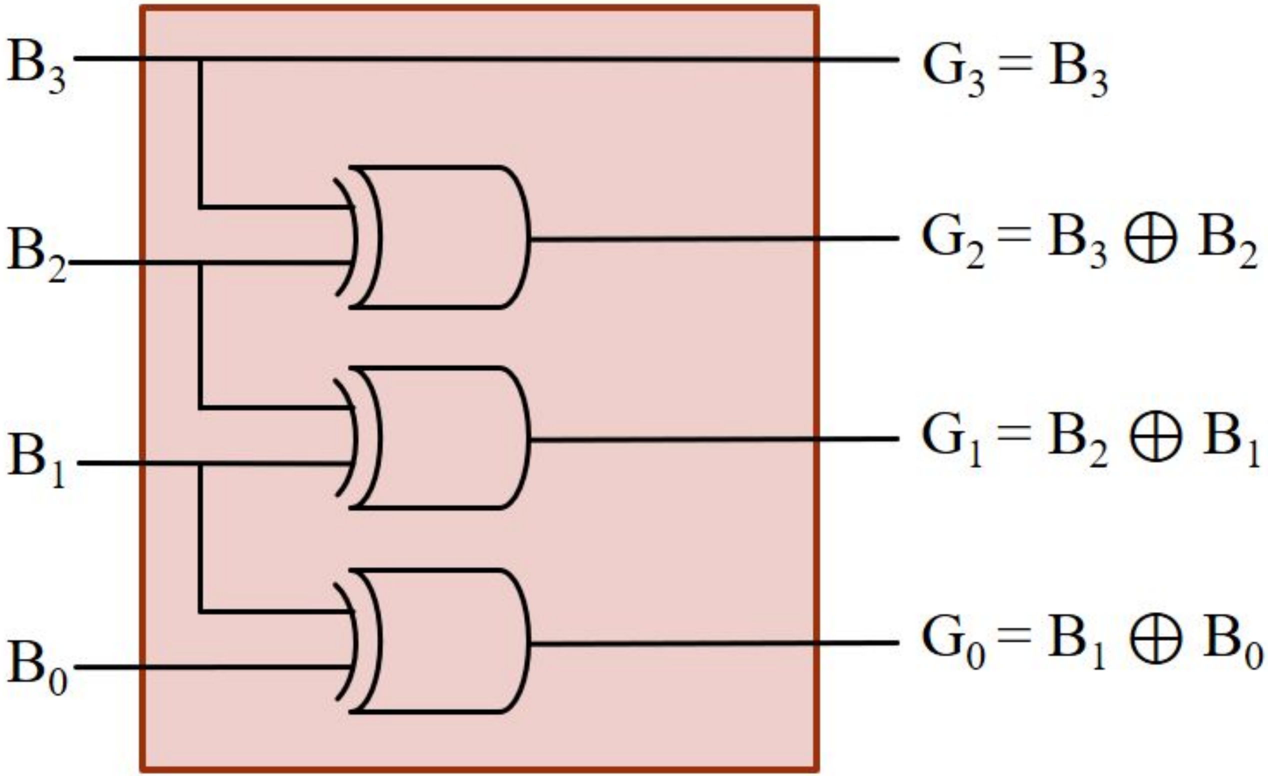
$$\boxed{G_1 = B_2 \oplus B_1}$$

B_3	B_2	B_1	B_0	G_0
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

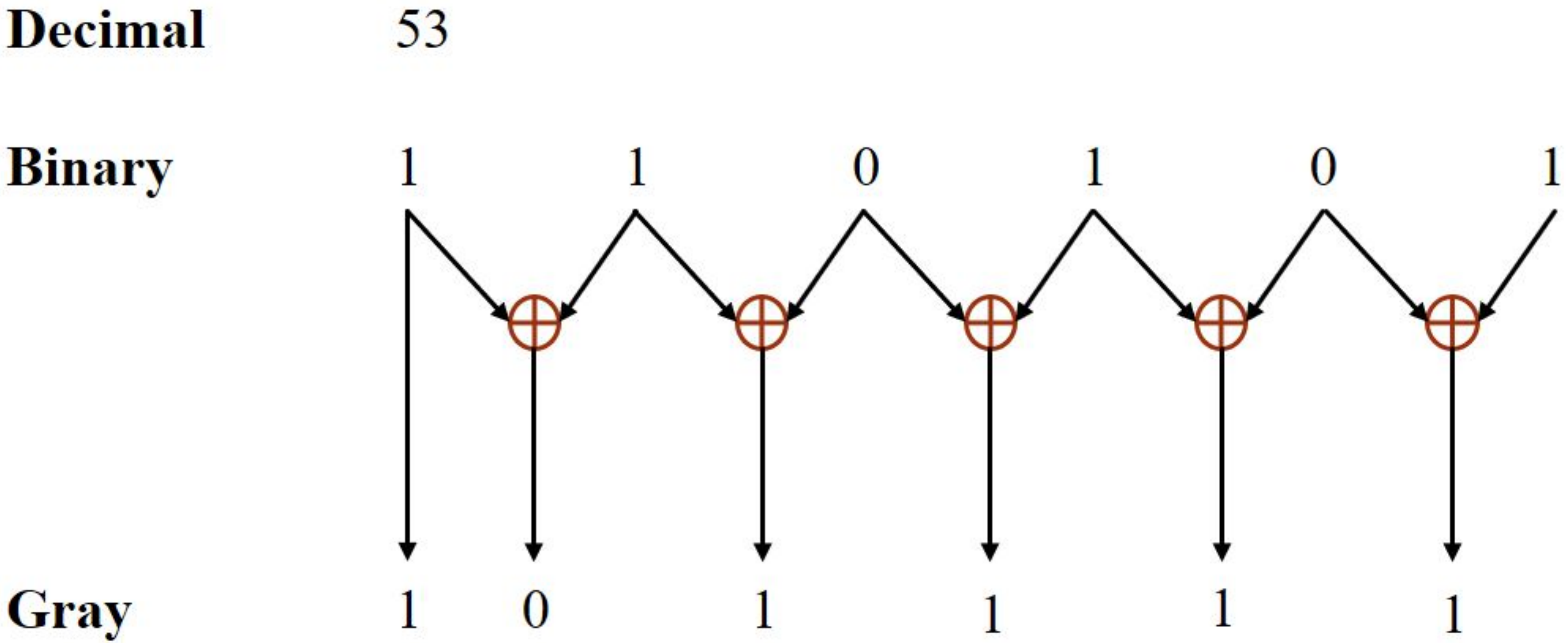
B_3B_2 \ B_1B_0	00	01	11	10
00		1		1
01		1		1
11		1		1
10		1		1

$$G_0 = B_1'B_0 + B_1B_0'$$

$$\boxed{G_0 = B_1 \oplus B_0}$$



4-Bit Binary to Gray Code Converter


Decimal
53
Binary
1 1 0 1 0 1
Gray
1 0 1 1 1 1

Gray to Binary Code Conversion

DECIMAL DIGIT	GRAY CODE				BINARY CODE			
	G_3	G_2	G_1	G_0	B_3	B_2	B_1	B_0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	1	0	0	1	0
3	0	0	1	0	0	0	1	1
4	0	1	1	0	0	1	0	0
5	0	1	1	1	0	1	0	1
6	0	1	0	1	0	1	1	0
7	0	1	0	0	0	1	1	1
8	1	1	0	0	1	0	0	0
9	1	1	0	1	1	0	0	1
10	1	1	1	1	1	0	1	0
11	1	1	1	0	1	0	1	1
12	1	0	1	0	1	1	0	0
13	1	0	1	1	1	1	0	1
14	1	0	0	1	1	1	1	0
15	1	0	0	0	1	1	1	1

G_3	G_2	G_1	G_0	B_3
0	0	0	0	0
0	0	0	1	0
0	0	1	1	0
0	0	1	0	0
0	1	1	0	0
0	1	1	1	0
0	1	0	1	0
0	1	0	0	0
1	1	0	0	1
1	1	0	1	1
1	1	1	1	1
1	1	1	0	1
1	0	1	0	1
1	0	1	1	1
1	0	0	1	1
1	0	0	0	1

G_3G_2 \ G_1G_0	00	01	11	10
00				
01				
11	1	1	1	1
10	1	1	1	1

$$\boxed{B_3 = G_3}$$

G_3	G_2	G_1	G_0	B_2
0	0	0	0	0
0	0	0	1	0
0	0	1	1	0
0	0	1	0	0
0	1	1	0	1
0	1	1	1	1
0	1	0	1	1
0	1	0	0	1
1	1	0	0	0
1	1	0	1	0
1	1	1	1	0
1	1	1	0	0
1	0	1	0	1
1	0	1	1	1
1	0	0	1	1
1	0	0	0	1

G_3G_2 \ G_1G_0	00	01	11	10
00				
01	1	1	1	1
11				
10	1	1	1	1

$$B_2 = G_3G_2' + G_2G_3'$$

$$\boxed{\begin{aligned} B_2 &= G_3 \oplus G_2 \\ B_2 &= B_3 \oplus G_2 \end{aligned}}$$

G_3	G_2	G_1	G_0	B_1
0	0	0	0	0
0	0	0	1	0
0	0	1	1	1
0	0	1	0	1
0	1	1	0	0
0	1	1	1	0
0	1	0	1	1
0	1	0	0	1
1	1	0	0	0
1	1	0	1	0
1	1	1	1	1
1	1	1	0	1
1	0	1	0	0
1	0	1	1	0
1	0	0	1	1
1	0	0	0	1

G_3G_2 \ G_1G_0	00	01	11	10
00			1	1
01	1	1		
11			1	1
10	1	1		

$$B_1 = G_3'G_2'G_1 + G_3'G_2G_1' + G_3G_2G_1 + G_3G_2'G_1'$$

$$B_1 = G_3\,(G_2 \oplus G_1)' + G_3'\,(G_2 \oplus G_1)$$

$$\boxed{\begin{array}{l} \mathbf{B_1 = G_3 \oplus G_2 \oplus G_1} \\ \mathbf{B_1 = B_2 \oplus G_1} \end{array}}$$

G_3	G_2	G_1	G_0	B_0
0	0	0	0	0
0	0	0	1	1
0	0	1	1	0
0	0	1	0	1
0	1	1	0	0
0	1	1	1	1
0	1	0	1	0
0	1	0	0	1
1	1	0	0	0
1	1	0	1	1
1	1	1	1	0
1	1	1	0	1
1	0	1	0	0
1	0	1	1	1
1	0	0	1	0
1	0	0	0	1

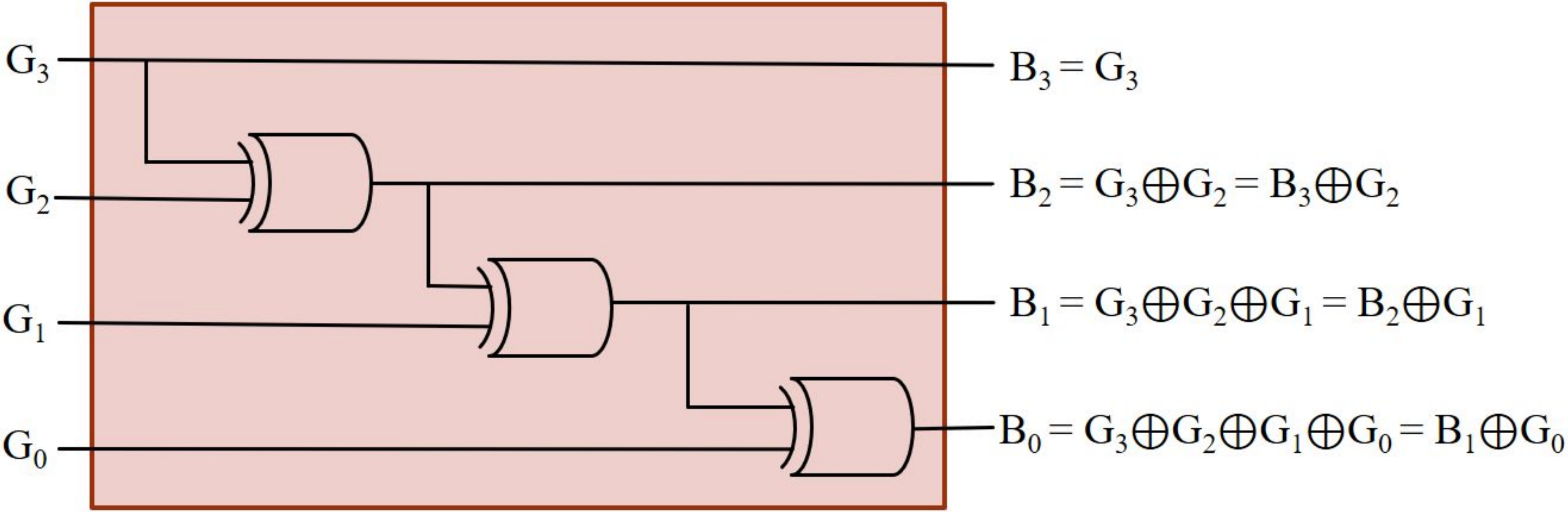
G_3G_2 \ G_1G_0	00	01	11	10
00		1		1
01	1		1	
11		1		1
10	1		1	

$B_0 = G_3'G_2'G_1'G_0 + G_3'G_2'G_1G_0' + G_3'G_2G_1'G_0' + G_3'G_2G_1G_0 + G_3G_2G_1'G_0 + G_3G_2G_1G_0' + G_3G_2'G_1'G_0' + G_3G_2'G_1G_0$

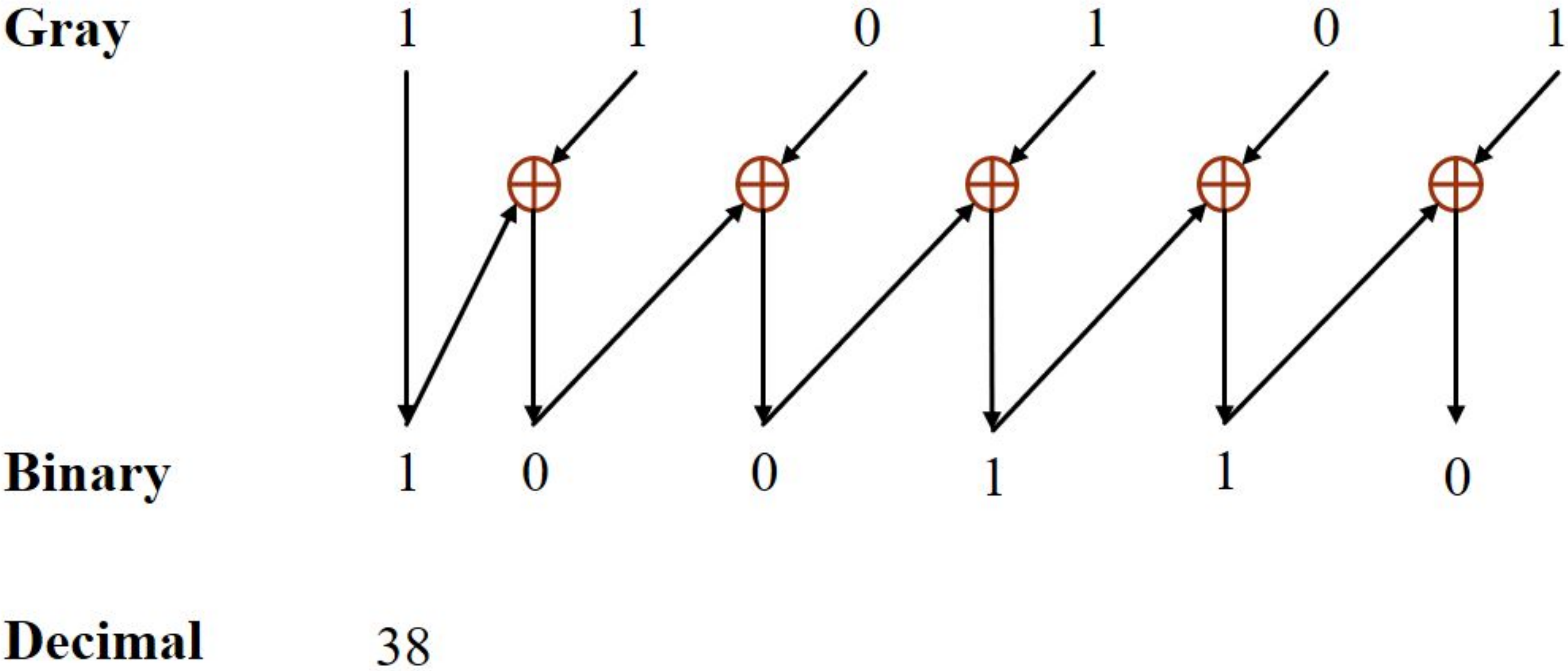
$B_0 = G_3'G_2'(G_1 \oplus G_0) + G_3'G_2(G_1 \oplus G_0)' + G_3G_2(G_1 \oplus G_0) + G_3G_2'(G_1 \oplus G_0)'$

$B_0 = (G_3 \oplus G_2)'(G_1 \oplus G_0) + (G_3 \oplus G_2)(G_1 \oplus G_0)'$

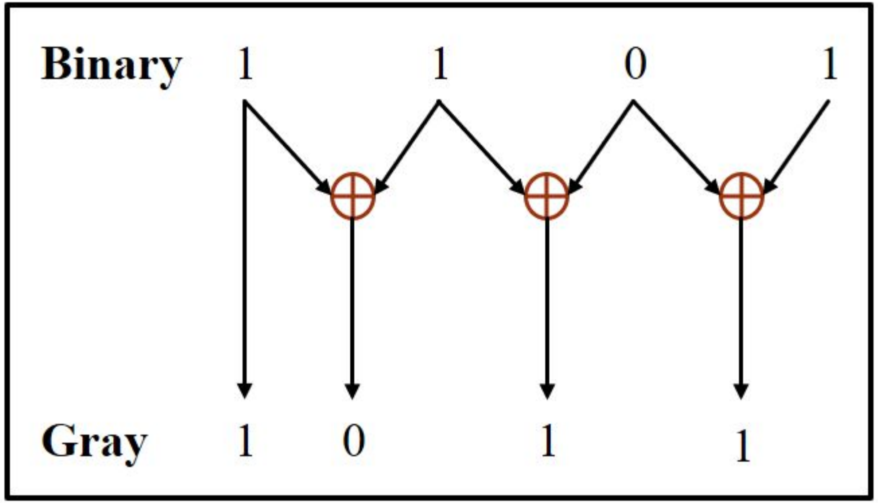
$$\boxed{\begin{array}{l} B_0 = G_3 \oplus G_2 \oplus G_1 \oplus G_0 \\ B_0 = B_1 \oplus G_0 \end{array}}$$



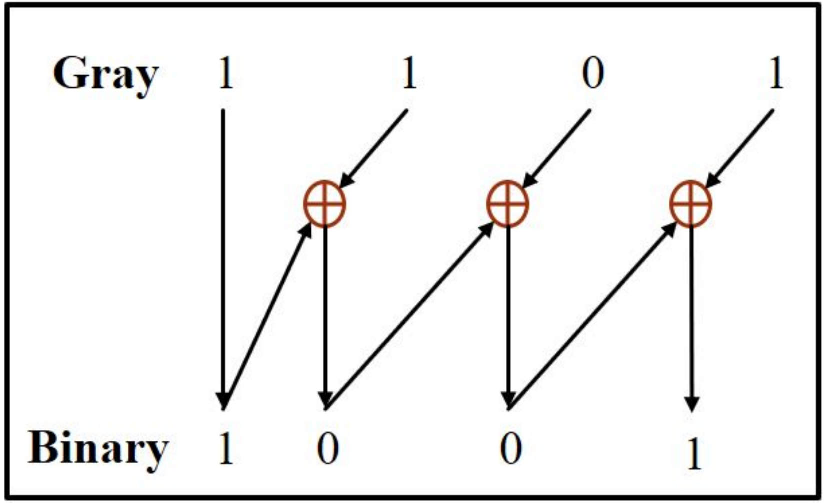
4-Bit Gray to Binary
Code Converter



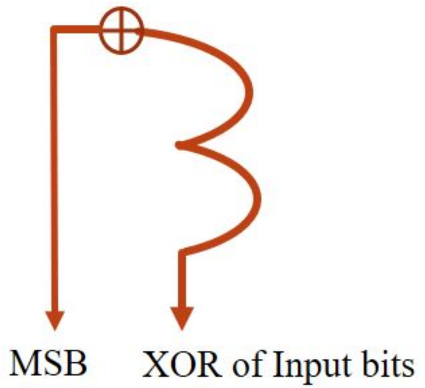
Gray
1 1 0 1 0 1
Binary
1 0 0 1 1 0
Decimal
38

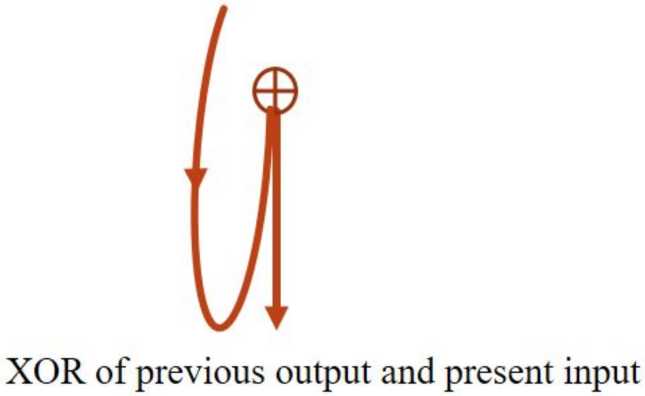


Binary to Gray Code Converter



Gray to Binary Code Converter





Excess 3 (XS-3) Codes

- 4-bit code
- BCD code + 011= XS-3 code
- Unweighted Code
- Self Complemented Code
- Mainly used for BCD subtraction
- Six invalid states: 0000, 0001, 0010, 1101, 1110, 1111

Excess-3 Code

Binary Code Decimal + 0011 = Excess-3 Code (XS-3)

	BCD	XS-3
0:	0000	0011
1:	0001	0100
2:	0010	0101
3:	0011	0110
4:	0100	0111
5:	0101	1000
6:	0110	1001
7:	0111	1010
8:	1000	1011
9:	1001	1100

53

1000 0110

$(6.9)_{10}$
$= (0110.1001)_{BCD}$
$= (1001.1100)_{XS\text{-}3}$

Decimal	XS-3
10	0100 0011
11	0100 0100
99	1100 1100
100	0100 0011 0011
101	0100 0011 0100
487	0111 1011 1010
...	...

Code Conversion:

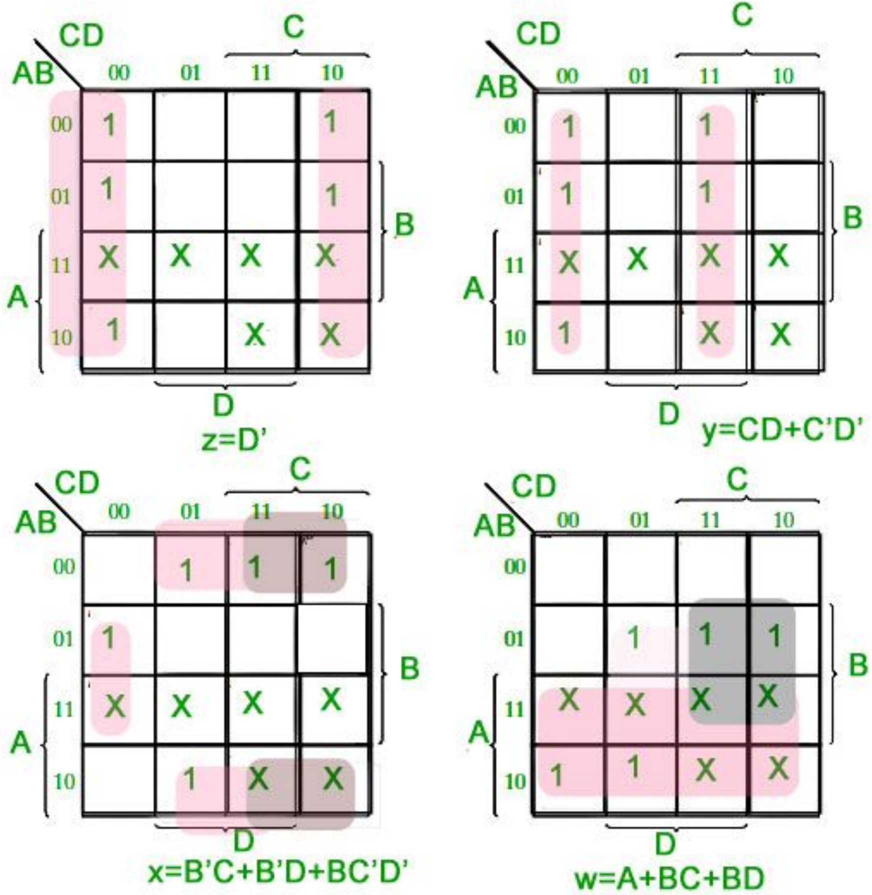
Approach 1:

- BCD to XS-3: Represent each XS-3 bit, $X_i = F(B_3, B_2, B_1, B_0)$
- XS-3 to BCD: $B_i = F(X_3, X_2, X_1, X_0)$ (Consider 'don't care' for unused input combinations.)

Approach 2:

- BCD to XS-3: Add 0011
- XS-3 to BCD: Subtract 0011 (Consider 4-bit Adder-Subtractor)

BCD(8421)				Excess-3			
A	B	C	D	w	x	y	z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0
1	0	1	0	X	X	X	X
1	0	1	1	X	X	X	X
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X



Excess-3				BCD			
w	x	y	z	A	B	C	D
0	0	0	0	X	X	X	X
0	0	0	1	X	X	X	X
0	0	1	0	X	X	X	X
0	0	1	1	0	0	0	0
0	1	0	0	0	0	0	1
0	1	0	1	0	0	1	0
0	1	1	0	0	0	1	1
0	1	1	1	0	1	0	0
1	0	0	0	0	1	0	1
1	0	0	1	0	1	1	0
1	0	1	0	0	1	1	1
1	0	1	1	1	0	0	0
1	1	0	0	1	0	0	1
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

wx \ yz	00	01	11	10
00	X	X	0	X
01	1	0	0	1
11	1	X	X	X
10	1	0	0	1

$D = z'$

wx \ yz	00	01	11	10
00	X	X	0	X
01	0	1	0	1
11	0	X	X	X
10	0	1	0	1

$C = y'z + yz'$

wx \ yz	00	01	11	10
00	X	X	0	X
01	0	0	1	0
11	0	X	X	X
10	1	1	0	1

$B = x'y' + x'z' + xyz$

wx \ yz	00	01	11	10
00	X	X	0	X
01	0	0	0	0
11	1	X	X	X
10	0	0	1	0

$A = wx + wyz$